

PCB Design & Hardware Prototyping

Go from schematics and breadboards to manufacturable PCB designs and practical hardware projects.

Recommended learning order

1. Circuit and schematic fundamentals
2. PCB layout basics
3. Traces, vias and stack-up
4. Manufacturing / assembly considerations
5. Prototype → test → debug → revise

Direct YouTube resources

Source	What to use it for	Open
Phil's Lab	PCB Design & Altium tutorials — beginner-to-advanced hardware design material.	YouTube →
Phil's Lab	Phil's Lab playlists — broader PCB, embedded and hardware-design material.	YouTube →
Paul McWhorter	Arduino tutorials for hands-on prototyping and circuit experiments.	YouTube →
Core Electronics	Raspberry Pi workshop for practical prototyping.	YouTube →

Tip: open the YouTube → buttons directly from the PDF. This roadmap is based on the supplied Hardware Learning Roadmap, with Computer Networks added as a separate course because you specifically requested it.

Prepared as a separate topic-wise learning PDF.